Jack Hollis Farrell

Department of Physics · Center for Theory of Quantum Matter
University of Colorado Boulder

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Education

Ph.D. Physics, University of Colorado Boulder — Expected 2027
- Advisor: Andrew Lucas

M.S. Physics, University of Colorado Boulder — 2025

Honours B.Sc. Physics, University of Toronto — 2017–2021
- Minors in Mathematics and English.

Research

Publications and Preprints

1. **Jack H. Farrell** and Andrew Lucas. "Characterizing electronic scattering rates with transport in multiterminal devices", arXiv: 2605.03030 (2026).
2. Ludwig Holleis, Youngjoon Choi, Canxun Zhang, **Jack H. Farrell**, Gabriel Bargas, Audrey Hsu, Zexing Chen, Ian Sackin, Wenjie Zhou, Yi Guo, Thibault Charpentier, Yifan Jiang, Benjamin A. Foutty, Aidan Keough, Martin E. Huber, Takashi Taniguchi, Kenji Watanabe, Andrew Lucas, and Andrea F. Young. "Cryogenic shock exfoliation for ultrahigh mobility rhombohedral graphite nanoelectronics", Nat. Nanotechnol. (2026).
3. Canxun Zhang, Evgeny Redekop, Hari Stoyanov, **Jack H. Farrell**, Sunghoon Kim, Ludwig Holleis, David Gong, Yongjoon Choi, Takashi Taniguchi, Kenji Watanabe, Martin E. Huber, Ania C. Bleszynski Jayich, Andrew Lucas, and Andrea F. Young. "Imaging electron hydrodynamics in a graphene flat band system", arXiv: 2603.11175 (2026).
4. Johannes Geurs, Tatiana A. Webb, Yinjie Guo, Itai Keren, **Jack H. Farrell**, Jikai Xu, Kenji Watanabe, Takashi Taniguchi, Dmitri N. Basov, James Hone, Andrew Lucas, Abhay Pasupathy, Cory R. Dean. "Supersonic flow and hydraulic jump in an electronic de Laval nozzle", arXiv: 2509.16321 (2025).
5. Xiaoyang Huang, **Jack H. Farrell**, Aaron J. Friedman, Isabella Zane, Paolo Glorioso, and Andrew Lucas. "Generalized time-reversal symmetry and effective theories for nonequilibrium matter", arXiv: 2310.12233 (2023).
6. **Jack H. Farrell**, Xiaoyang Huang, and Andrew Lucas. "Hydrodynamics with helical symmetry", arXiv: 2208.12269 (2022).
7. **Jack H. Farrell**, Nicolas Grisouard, and Thomas Scaffidi. "Terahertz radiation from the Dyakonov-Shur instability of hydrodynamic electrons in a Corbino geometry". Phys. Rev. B 106, 195432 (2022).

Manuscripts in Preparation

1. **Jack H. Farrell**, Akshay Pal, and Andrew Lucas. "Hydrodynamic-to-ballistic crossover of electron transport in complex devices". Manuscript in preparation (2026).
2. Akshay Pal, Andrew Hicks, Andrew Lucas, and **Jack H. Farrell**. "Electron hydrodynamics with an open Fermi surface". Manuscript in preparation (2026).

3. Elijah G. Lew-Smith, Marvin Qi, **Jack H. Farrell**, and Andrew Lucas. "A lattice model of constant density flocking without long range order". Manuscript in preparation (2026).
4. Zachary Hill, Andrew Lucas, and **Jack H. Farrell**. "Nonlinear kinetic theory of interacting electrons in confined geometries". Manuscript in preparation (2026).

Presentations

Invited Talks

- Centre de Physique Théorique, École Polytechnique (2026). "Computational Fluid Mechanics for Electrons".
- Cory Dean Group Meeting, Columbia University (2025). "Compressible Flow Instabilities in Electronic Liquids".
- CTQM Seminar, Center for Theory of Quantum Matter (2024). "Generalized Time-Reversal Symmetry and Effective Theories for Nonequilibrium Matter".

Contributed Talks

- Correlated Quantum Systems Workshop, Institut Pascal (2026). "2D Electron Transport in Complex Geometries". (video)
- APS Global Physics Summit (2026). "2D Electron Transport through Complex Geometries".
- APS March Meeting (2025). "Endpoint of a Ballistic Dyakonov-Shur Instability".
- Quantum, Classical, and Active Fluids Workshop, CUNY Graduate Center (2024). "Constant Density Flocking without Long-Range Order". (video)
- APS March Meeting (2024). "Effective Theories for Nonequilibrium Matter".
- APS March Meeting (2023). "Hydrodynamics with Helical Symmetry".

Poster Presentations

- UQM: Ultra Quantum Matter (2024). "Generalized Time-Reversal Symmetry and Effective Theories for Nonequilibrium Matter".

Research Mentoring

Primary research contact for the following students:

- Julia Gao, Boulder High School — 2026–present
- Zachary Hill, UT Austin REU — 2026–present
- Andrew Hicks, CU Boulder, undergraduate (now Ph.D. student, Yale University) — 2022–2024
- Elijah G. Lew-Smith, Brown U. REU (now Ph.D. student, MIT) — 2023–2024

Teaching

Instructor of Record, University of Colorado Boulder

- PHYS 1400: Fundamentals of Scientific Inquiry — 2025

Guest Lecturer, University of Colorado Boulder

- PHYS 7810: Graduate Special Topics: Hydrodynamics — 2026
- PHYS 5210: Graduate Theoretical Mechanics — 2024

- PHYS 4410: Quantum Mechanics II 2023

Teaching Assistant, University of Colorado Boulder
- PHYS 7810: Graduate Special Topics in Physics: Hydrodynamics Spring 2024
- PHYS 3090: Introduction to Quantum Computing Spring 2024
- PHYS 1125: General Physics II for Majors Spring 2023
- PHYS 1110: General Physics I Fall 2023
- PHYS 1110: General Physics I Fall 2022
- PHYS 1110: General Physics I Spring 2022
- PHYS 1110: General Physics I Fall 2021

Teaching Assistant, University of Toronto
- Python programming support for physics majors 2020–2021

Honours

Department of Physics Supplemental Fellowship, University of Colorado Boulder 2026

NSERC Canada Graduate Scholarship (CGS-M), NSERC 2021
- Awarded; declined.

University of Toronto National Scholarship 2017–2021
- Full scholarship to U of T, covering tuition and living fees for four years.

NSERC Undergraduate Summer Research Award, University of Toronto 2020

Hymie and Roslyn Mida Award in Theoretical Physics, University of Toronto 2020

NSERC Undergraduate Summer Research Award, University of Toronto 2019

University of Toronto Scholar 2018–2019

Service

Program Assistant, Boulder Summer School for Condensed Matter and Materials Physics
- Geometry and Topology in Soft Matter 2026
- Dynamics of Strongly Correlated Electrons 2025
- Self-Organizing Matter: From Inanimate to the Animate 2024
- Non-Equilibrium Quantum Dynamics 2023
- Hydrodynamics Across Scales 2022